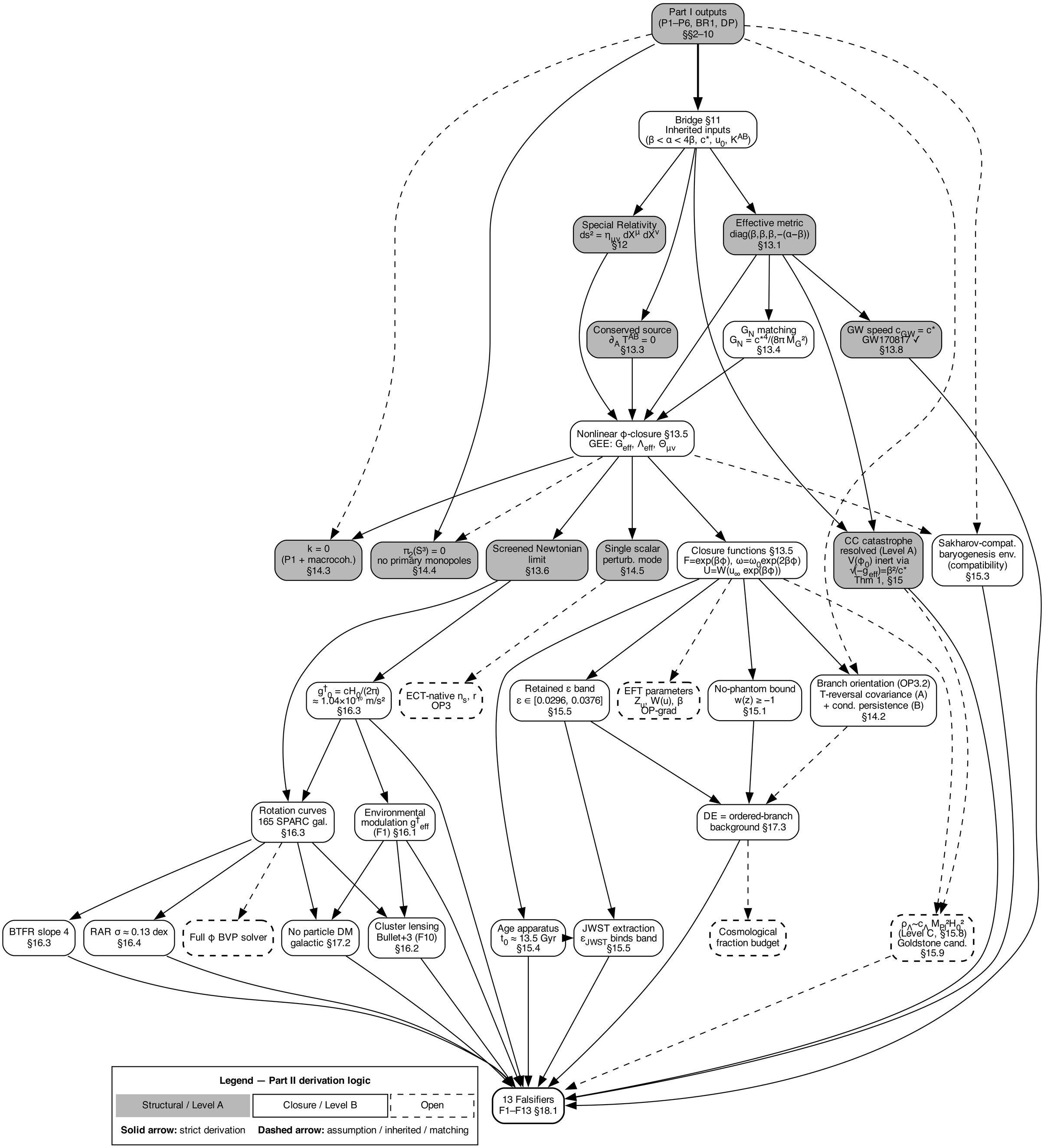
Part I outputs
(P1–P6, BR1, DP)
§§2–10
Bridge §11
Inherited inputs
(β < α < 4β, c*, u0, K^AB)
Special Relativity
ds² = ηMν dXμ dXν
§12
Effective metric
diag(β,β,β,−(α−β))
§13.1
Conserved source
∂A TAB = 0
§13.3
GN matching
GN = c*4/(8π MG²)
§13.4
GW speed cGW = c*
GW170817 ✓
§13.8
Nonlinear ϕ-closure §13.5
GEE: Geff, Λeff, Θμν
k = 0
(P1 + macrocoh.)
§14.3
π2(S³) = 0
no primary monopoles
§14.4
Screened Newtonian
limit
§13.6
Single scalar
perturb. mode
§14.5
Closure functions §13.5
F=exp(βϕ), ω=ω0exp(2βϕ)
U=W(u∞ exp(βϕ))
CC catastrophe
resolved (Level A)
V(ϕ0) inert via
√(−geff)=β²/c*
Thm 1, §15
Sakharov-compat.
baryogenesis env.
(compatibility)
§15.3
g†0 = cH0/(2π)
≈ 1.04×10⁻¹⁰ m/s²
§16.3
ECT-native ns, r
OP3
Retained ε band
ε ∈ [0.0296, 0.0376]
§15.5
EFT parameters
Zu, W(u), β
OP-grad
No-phantom bound
w(z) ≥ −1
§15.1
Branch orientation (OP3.2)
T-reversal covariance (A)
+ cond. persistence (B)
§14.2
Rotation curves
165 SPARC gal.
§16.3
Environmental
modulation g†eff
(F1) §16.1
DE = ordered-branch
background §17.3
BTFR slope 4
§16.3
RAR σ ≈ 0.13 dex
§16.4
Full ϕ BVP solver
No particle DM
galactic §17.2
Cluster lensing
Bullet+3 (F10)
§16.2
Age apparatus
t0 ≈ 13.5 Gyr
§15.4
JWST extraction
εJWST binds band
§15.5
Cosmological
fraction budget
ρΛ~cΛ MPl²H0²
(Level C, §15.8)
Goldstone cand.
§15.9
13 Falsifiers
F1–F13 §18.1
Legend — Part II derivation logic
Structural / Level A
Closure / Level B
Open
Solid arrow: strict derivation
Dashed arrow: assumption / inherited / matching